

BCA SEMESTER V • UNIT IV: MULTIVARIATE METHODS

Multivariate Normal Distribution, Classification & Tuning Complexity

Prescribed Textbook: Ethem Alpaydın, *Introduction to Machine Learning* (2004, PHI),
Chapter 5

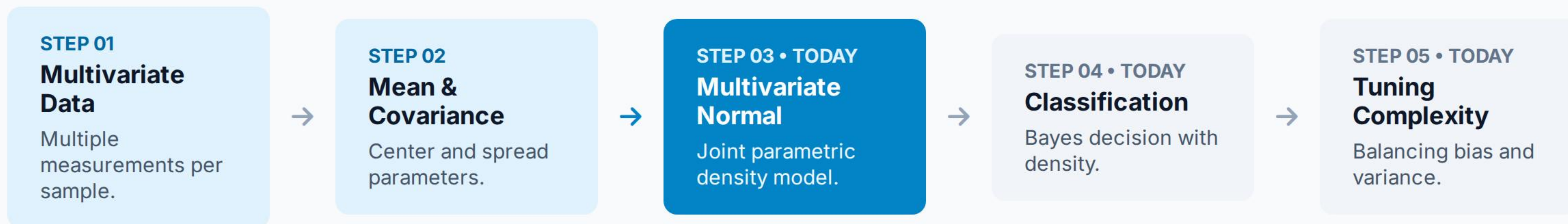


Classroom Lecture Series



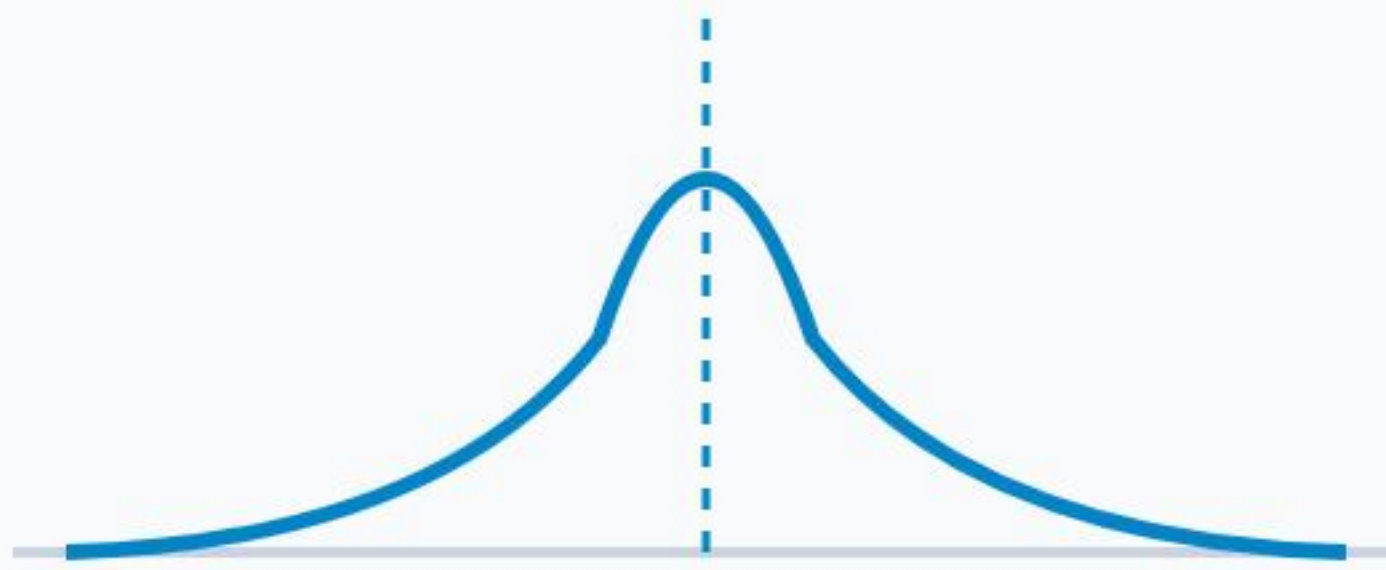
Sections 5.4 – 5.6

Connection With Previous Class



Class Goal: Moving from sample parameter estimates to predictive probabilistic models and decision rules.

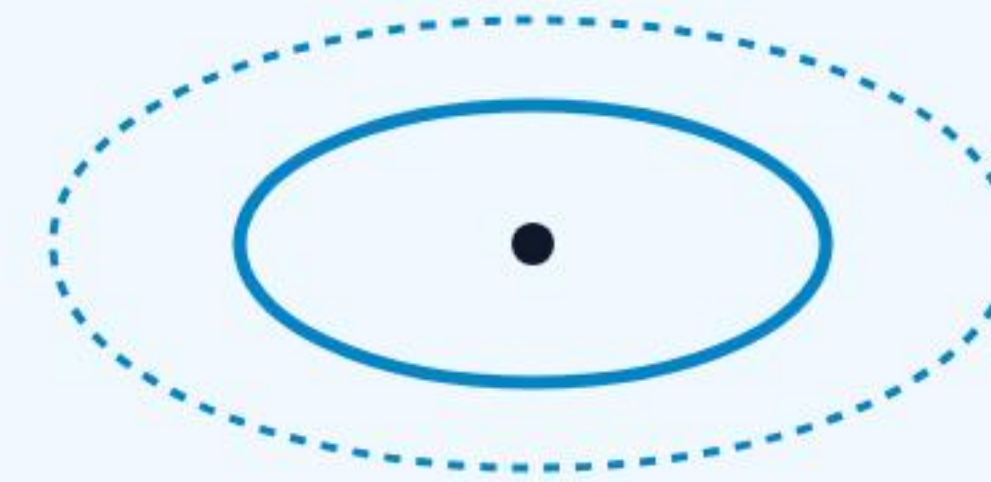
Univariate to Multivariate Normal



Univariate Normal: $N(\mu, \sigma^2)$

Models a single continuous feature.

Mean μ (scalar) • Variance σ^2 (scalar)



Multivariate Normal: $N_d(\mu, \Sigma)$

Models d features simultaneously.

Mean vector μ • Covariance matrix Σ

Multivariate Normal Distribution

Textbook Standard Notation:

$$\mathbf{x} \sim N_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$\mathbf{x}$

Observation Vector

Column vector of d feature values.

d

Dimensionality

Total number of measured input features.

$\boldsymbol{\mu}$

Mean Vector

Vector holding center coordinates ($d \times 1$).

$\boldsymbol{\Sigma}$

Covariance Matrix

Spread and pairwise relations ($d \times d$).

The Mean Vector: Distribution Center

Defines Central Location

The mean vector μ stacks expected values of each individual feature into a single column matrix.

Peak density of the multivariate bell occurs exactly at $x = \mu$.

SIMPLE CLASSROOM EXAMPLE (D = 2)

Feature 1
Height (cm)

Feature 2
Weight (kg)

$$\mu = [168.5 , 64.2]^T$$

The Covariance Matrix: Spread & Correlation

2 × 2 Covariance Matrix

$$\Sigma = \begin{vmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{vmatrix}$$

Symmetric matrix: $\sigma_{12} = \sigma_{21}$

Diagonal Elements (σ_1^2, σ_2^2)

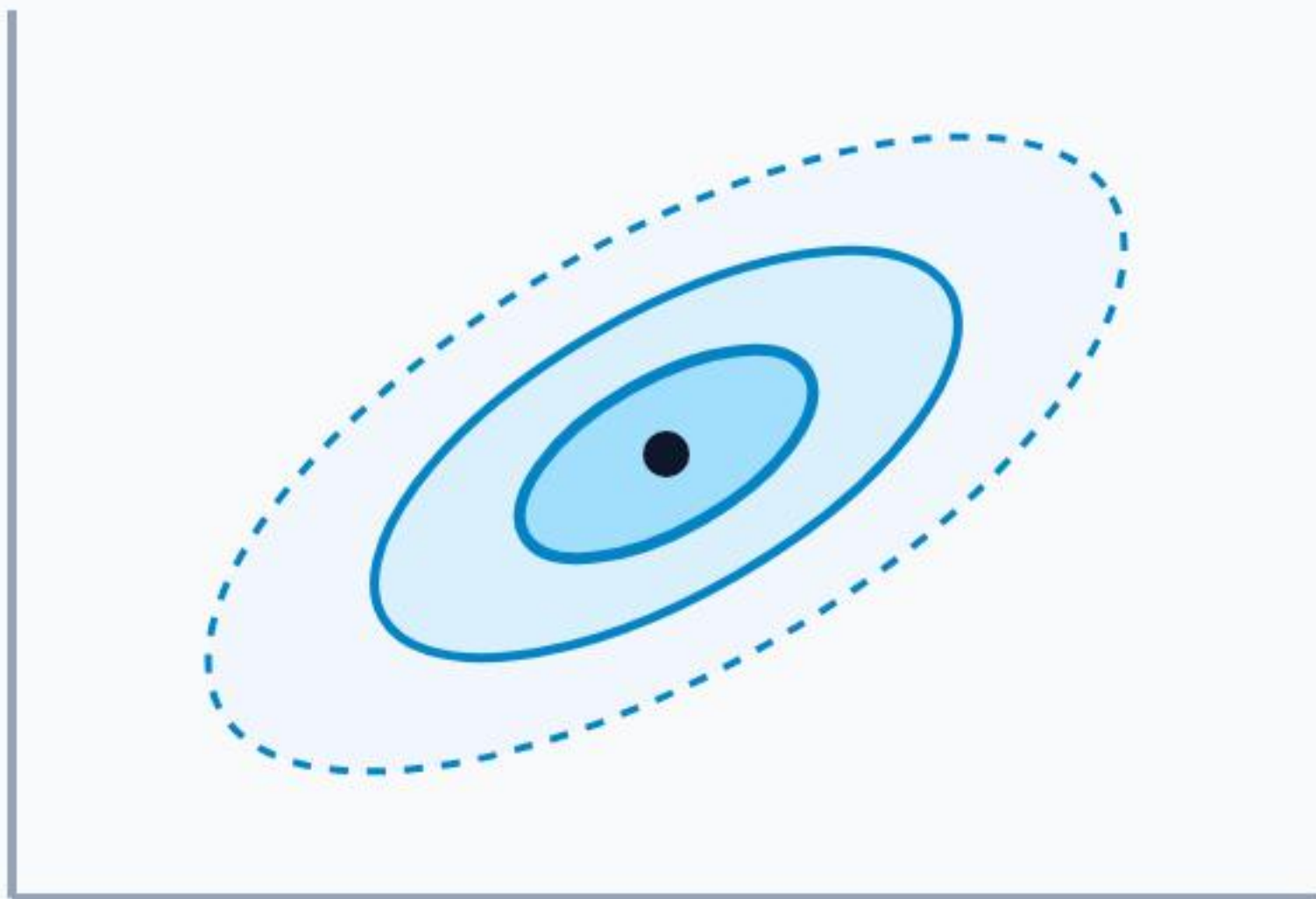
Individual feature variance along coordinate axes.

Off-Diagonal Elements (σ_{12}, σ_{21})

Linear association between feature pairs.

Non-zero covariance tilts distribution ellipses.

2D Geometry & Distribution Shape



1. Center Point

Positioned exactly at mean vector μ .

2. Contour Spread

Length of axes proportional to variances.

3. Tilt / Orientation

Angle reflects sign and magnitude of covariance.

Multivariate Density Function

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right]$$

$|\Sigma|^{1/2}$

Determinant scales volume of dispersion.

Σ^{-1}

Inverse covariance normalizes correlations.

$(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$

Mahalanobis squared distance from center.

$d / 2$

Normalizing constant for d dimensions.

Why Use Multivariate Normal?



Joint Modeling

Models features together without assuming artificial independence.



Mathematical Elegance

Fully specified by two parameters: mean vector and covariance matrix.



Real-World Relevance

Continuous real-world sensor measurements cluster naturally around means.

Toward Multivariate Classification

The Central Question

If data from different classes follow multivariate normal distributions, how do we assign a new observation?

1. Training Data

Group points by class label.



2. Parameter Fit

Estimate μ_i and Σ_i per class.



3. Density Scoring

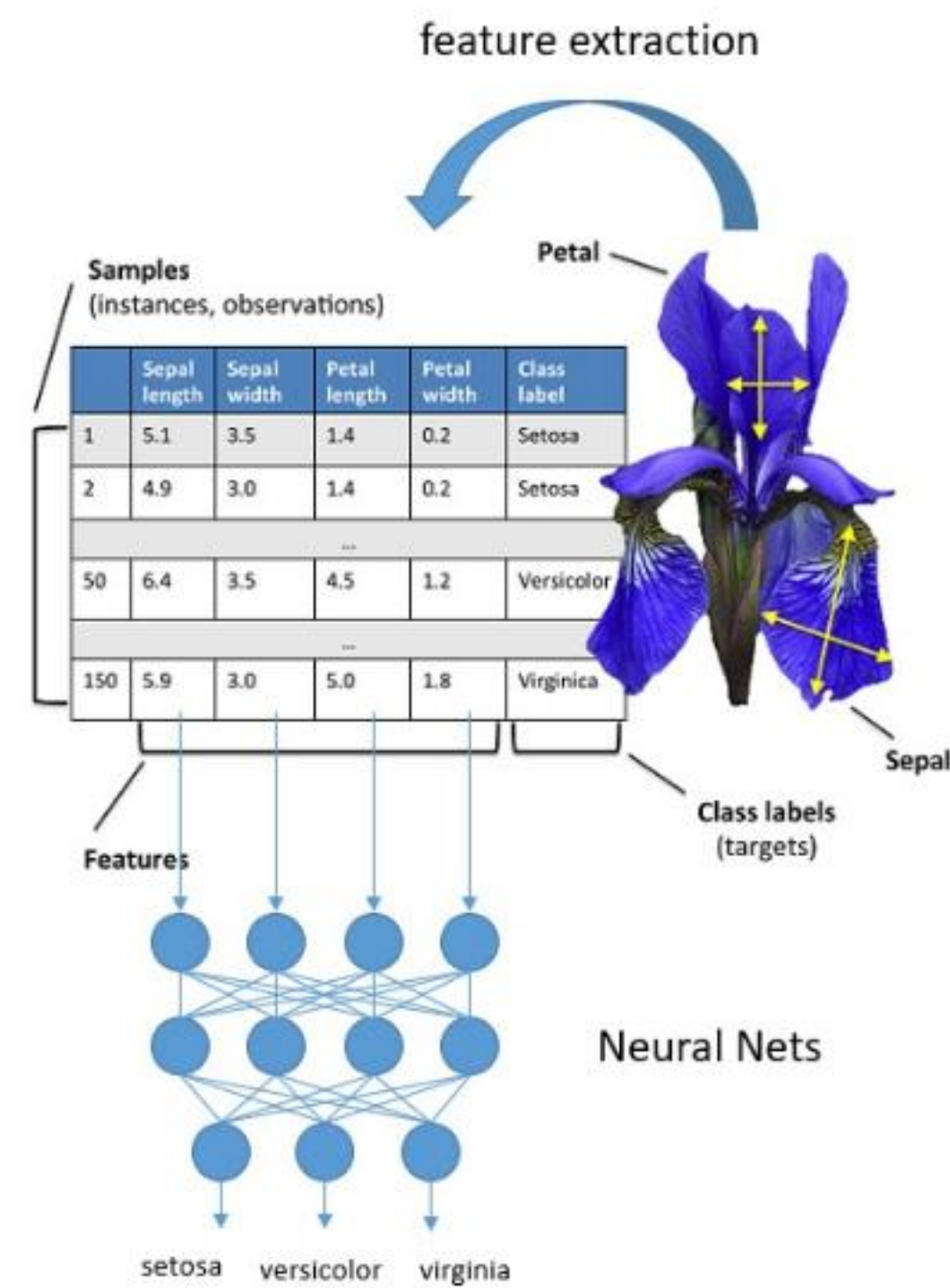
Compute $p(x \mid C_i)$.



4. Classification

Assign to highest likelihood.

Multivariate Classification



Definition

Classifying instances using multiple simultaneous features.

Classic Textbook Example ($d = 4$)

Iris flowers: Sepal Length, Sepal Width, Petal Length, Petal Width.

Why Multiple Features?

Separate overlapping classes through joint multidimensional space.

Class-Conditional Gaussian Model

Each class C_i produces feature vectors according to its own multivariate normal density:

CLASS 1 (C_1)

$$\mathbf{x} \mid C_1 \sim N_d(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$$

- $\boldsymbol{\mu}_1$: Center of Class 1 instances
- $\boldsymbol{\Sigma}_1$: Dispersion of Class 1 instances

CLASS 2 (C_2)

$$\mathbf{x} \mid C_2 \sim N_d(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$$

- $\boldsymbol{\mu}_2$: Center of Class 2 instances
- $\boldsymbol{\Sigma}_2$: Dispersion of Class 2 instances

6-Step Classification Flow

STEP 1

Training Data

Collect labeled sample vectors.

STEP 2

Separate Classes

Partition data by class label C_i .

STEP 3

Estimate Means

Compute sample mean vectors m_i .

STEP 4

Estimate Covariances

Compute sample matrices S_i .

STEP 5

Calculate Likelihoods

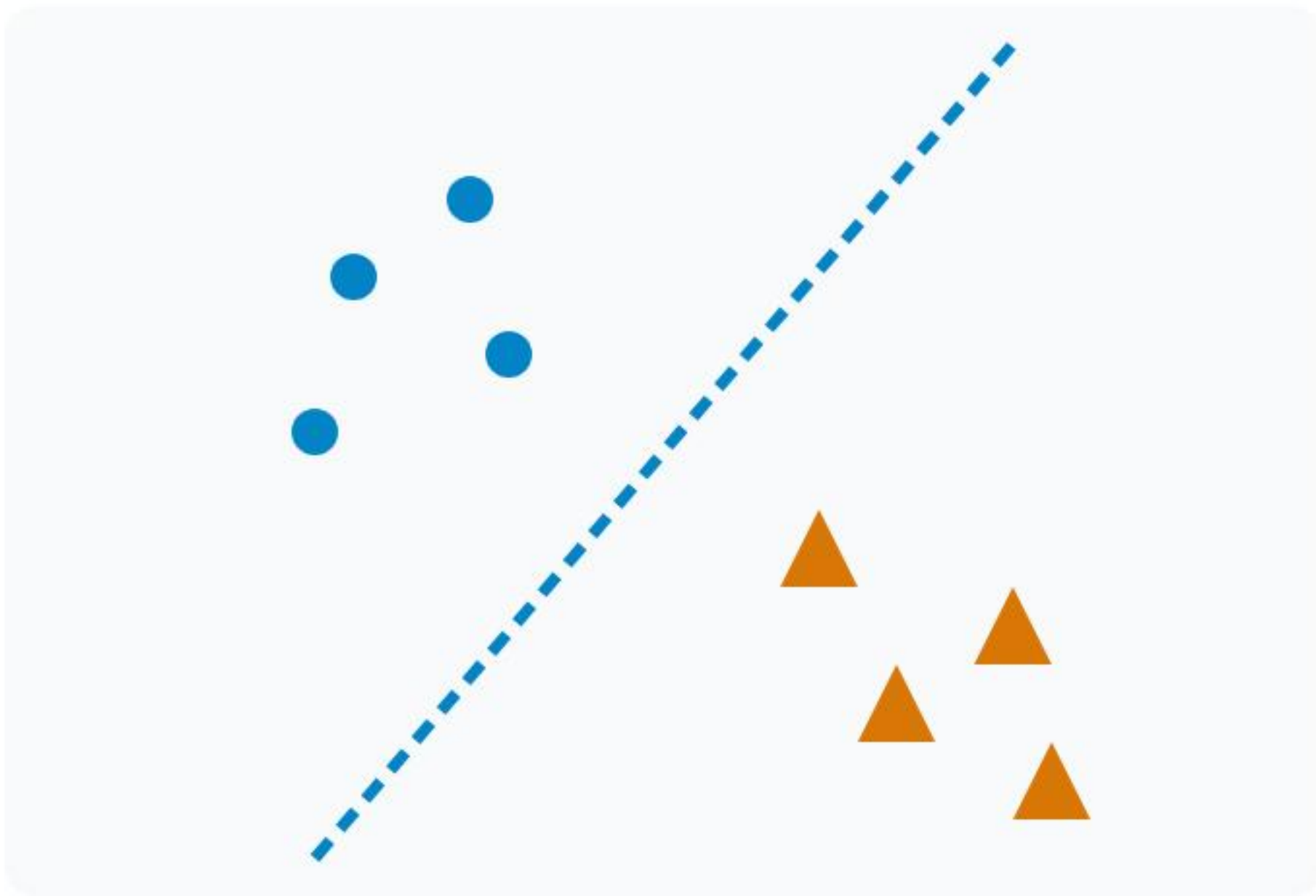
Evaluate $p(x \mid C_i)$ for input vector x .

STEP 6

Assign Class

Pick class with maximal discriminant.

Decision Boundaries



Definition

Locus of points where two classes have equal posterior probabilities.

Boundary Geometry

Linear when covariances are equal. Quadratic when covariances differ.

Model Complexity Dilemma

Simple Models

- Fewer parameters to estimate
- Robust on small datasets
- Low estimation variance

Risk: High Bias (underfitting)

Complex Models

- Many parameters to estimate
- Flexible decision boundaries
- Captures intricate patterns

Risk: High Variance (overfitting)

Simple vs Complex Covariance

COMMON COVARIANCE

Shared Matrix ($\Sigma_i = \Sigma$)

- All classes share identical orientation
- Generates **linear** decision boundaries

Fewer parameters: $d(d + 1)/2$ values

CLASS-SPECIFIC COVARIANCE

Arbitrary Matrices (Σ_i)

- Each class has distinct shape and tilt
- Generates **quadratic** decision boundaries

More parameters: $K \times d(d + 1)/2$ values

Complexity & Bias–Variance



Too Simple (High Bias)

Restricting covariance excessively cannot capture true correlation.



Too Complex (High Variance)

Full covariance with small data causes erratic estimates.



Optimal Tuning

Cross-validation balances bias and variance for minimal test error.

The Small Dataset Challenge

$O(d^2)$ Parameters

"More complex models require significantly more data."

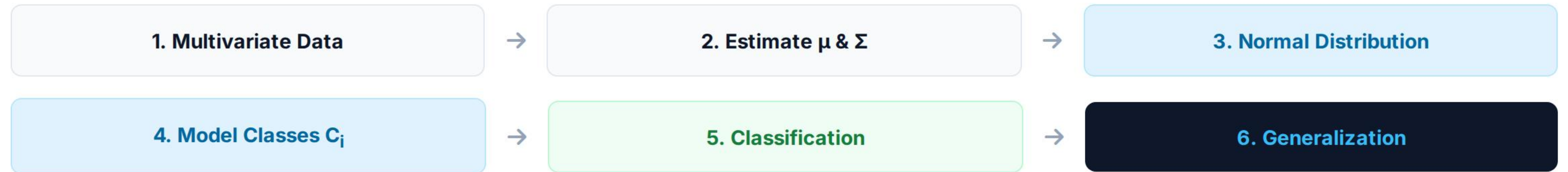
When Data Is Scarce

Sample covariance matrices become singular or non-invertible.

The Solution

Constrain covariance: share matrices across classes or force diagonal.

Complete Learning Architecture



Unit Review & Exam Questions

Short-Answer Questions (2 Marks)

1. Define multivariate normal distribution.
2. What is a mean vector?
3. Distinguish variance from covariance in matrix Σ .
4. What is a decision boundary?
5. What does tuning complexity mean?

Long-Answer Questions (5/10 Marks)

1. Explain the multivariate normal density with all symbols.
2. Detail the 6-step multivariate classification process.
3. Contrast shared covariance vs class-specific covariance.
4. Discuss the small dataset problem and bias-variance tradeoff.

Image Sources



<https://sefiks.com/wp-content/uploads/2018/08/neural-nets-for-iris.jpg>

Source: sefiks.com